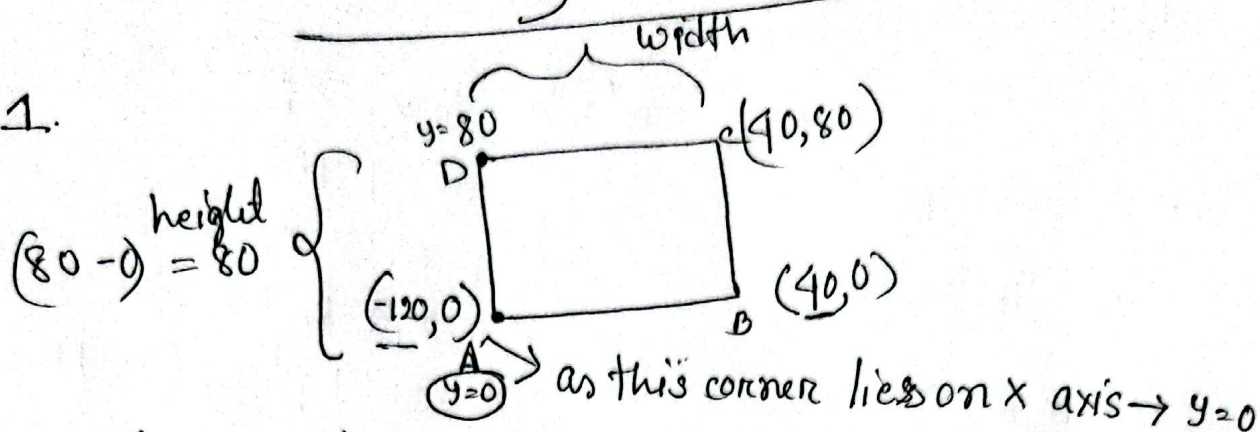


Quiz-2 [SET-A] solve

1.



here, width = 2 * height

$$\text{width} = 2 * 80$$

$$\text{width} = 160$$

$\therefore x$ of bottom left corner $\rightarrow (40 - 160) = -120$

$$\therefore x_{\min} = -120$$

$$x_{\max} = 40$$

$$y_{\min} = 0$$

$$y_{\max} = 80$$

$$A(-120, 0) \rightarrow 0100$$

$$B(40, 80) \rightarrow 1010$$

$$\text{AND} \rightarrow 0000$$

can be partially accepted

$$A \rightarrow \begin{array}{c} B \\ 0 \underline{1} 0 0 \\ \hline T \ B \ R \ L \end{array}$$

clipped point. $A'(-13, 0)$

$$\therefore A'(-13, 0) \rightarrow 0000$$

$$B(40, 80) \rightarrow 1010$$

$$\text{AND} \rightarrow 0000$$

can be partially accepted

As $B=1$ intersects Bottom boundary $\rightarrow y_{\min} \rightarrow y$ $y=0$
 $A'(x, y_{\min}) \rightarrow A'(x, 0)$

$$y - y_1 = m(x - x_1)$$

$$x = x_1 + \frac{1}{m}(y - y_1)$$

$$= -20 + \frac{1}{1.429}(0 + 10)$$

$$x = -13$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{80 - 0}{50 - (-20)}$$

$$= \frac{80}{70}$$

$$m = 1.1429$$

$$m = 1.429$$

here,

$$B(50, 90) \xrightarrow{T} 1010$$

clipped point B' .

$$B'(42.98, 80)$$

~~$B(50, 90)$~~

$$A' \rightarrow 0000$$

$$B' \rightarrow 0010$$

$$\begin{array}{r} \text{AND} \quad 0000 \\ \hline \end{array}$$

can be partially accepted

$$B' \xrightarrow{R} 0010$$

$$\therefore B''(40, 75.74)$$

$$A' \rightarrow 0000$$

$$B'' \rightarrow 0000$$

~~$B(50, 90)$~~
it is partially accepted.

clipped points are:

$$A'(-10, 0)$$

$$B''(40, 75.74)$$

$$B''(40, 76) \leftarrow (\text{round})$$

$T=1$, intersects top boundary

$$\therefore y = y_{\max}$$

$$B'(x, y_{\max}) \rightarrow B'(x, 80)$$

$$m = 1.429$$

$$y - y_1 = m(x - x_1)$$

$$x = x_1 + \frac{1}{m}(y - y_1)$$

$$x = -20 + \frac{1}{1.429}(80 + 10)$$

$$x = 42.98$$

$R=1$, intersects Right boundary.

$$x = x_{\max} = 40$$

$$\therefore B''(x_{\max}, y) \rightarrow B''(40, y)$$

$$m = 1.429$$

$$y - y_1 = m(x - x_1)$$

$$y = y_1 + m(x - x_1)$$

$$= -10 + 1.429(40 + 20)$$

$$y = 75.74$$